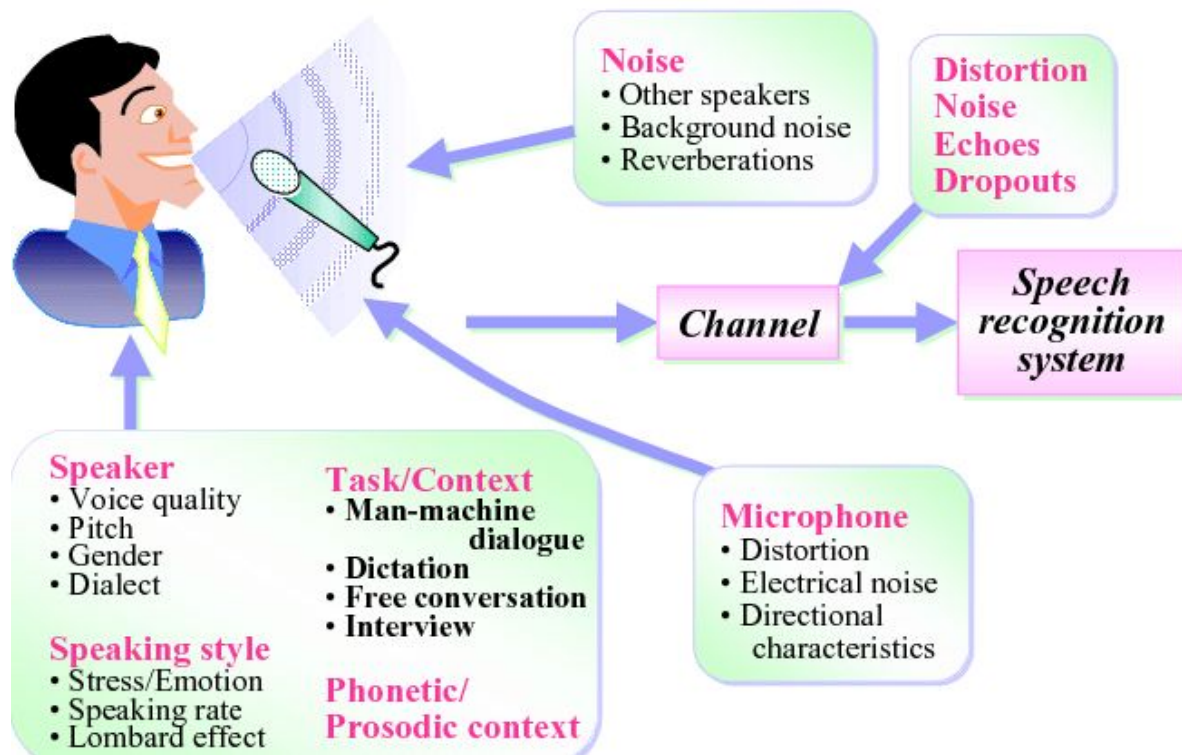


myMediWhisper: Construction of Burmese Medical Speech Corpus and Whisper Fine-Tuning for Clinical Dialogue ASR

Ye Kyaw Thu, Ye Bhone Lin, Thura Aung, Htet Arkar, Myat Oo Swe, Thet Htet San, Min Thiha Tun, Thazin Myint Oo, Thepchai Supnithi



Why Burmese Medical ASR?

Medical ASR Impact: Converts clinical dialogue into text, streamlining electronic health record (EHR) entry and reducing clinical workload.

Low-Resource Barrier: Burmese lacks open-access medical speech corpora, severely limiting domain-adapted ASR development.

Acoustic Challenges: Clinical environments exhibit high reverberation, room noise, and complex multi-speaker dialogue setups.

myMedi Whisper Goal: Build a high-quality 28-hour validated corpus and benchmark Whisper models under realistic room acoustics.

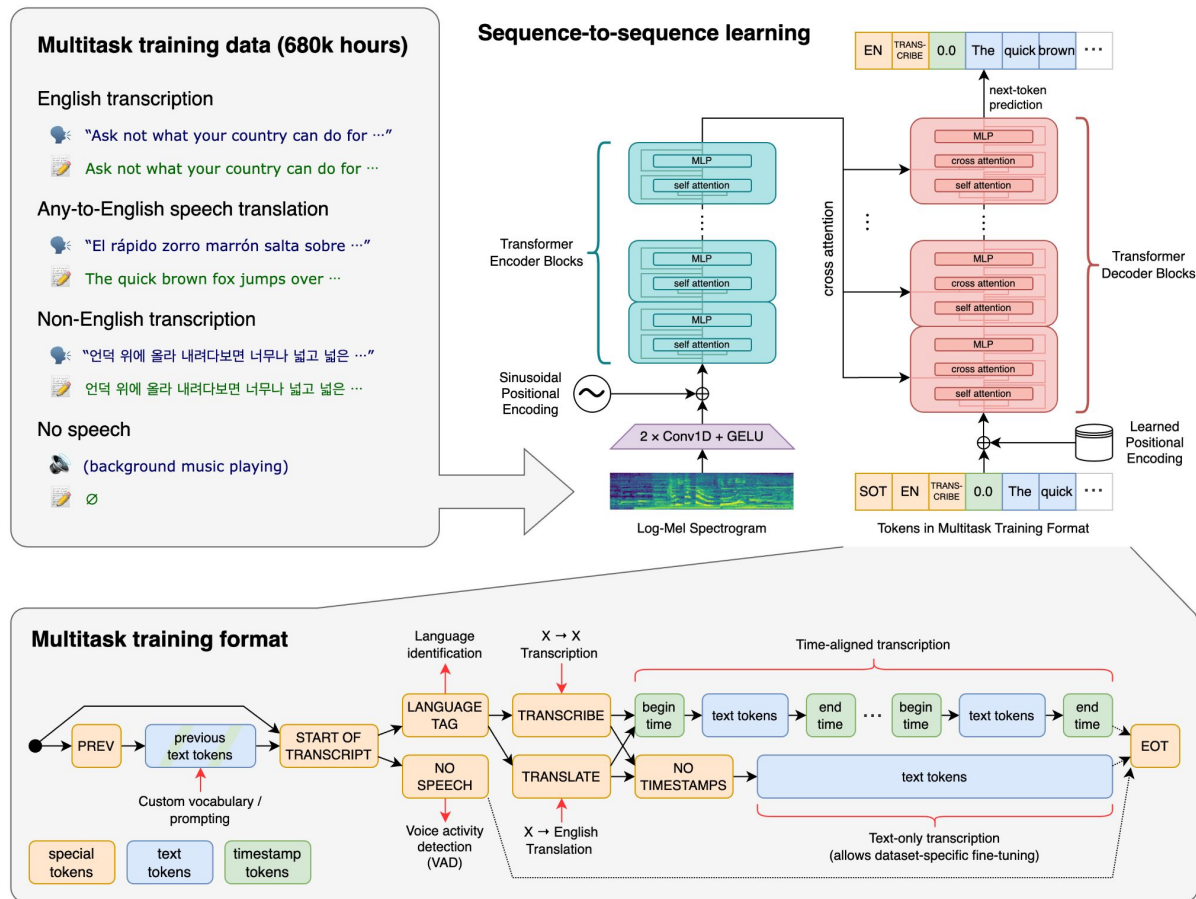
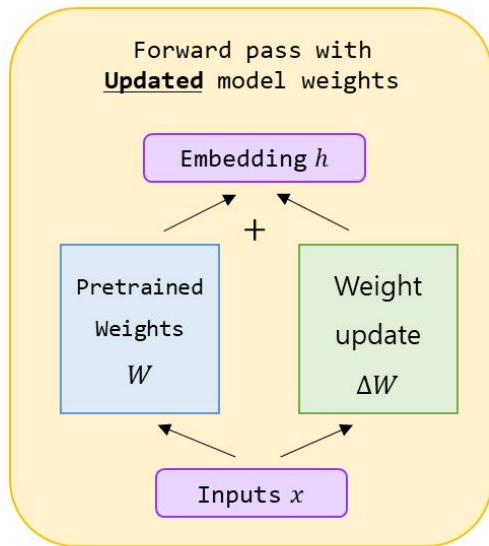
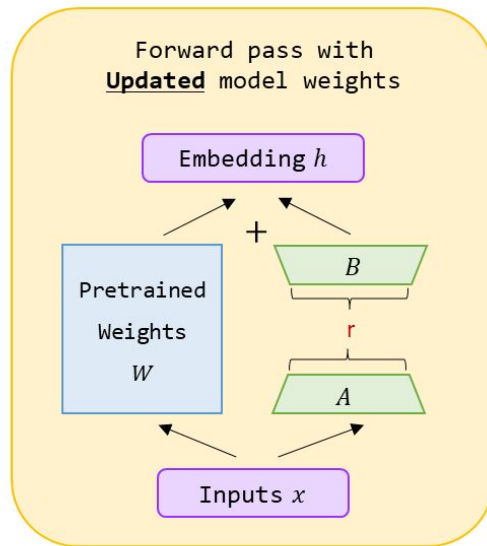


Fig. 2. Whisper Architecture (Alec Radford, 2022)

Regular finetuning



LoRA weights, A and B, represent ΔW



In our experiment:
Low-rank capacity = 128
 $\alpha = 256$
Dropout = 0.05

$$\mathbf{h} = \mathbf{W}_0\mathbf{x} + \Delta\mathbf{W}\mathbf{x} = \mathbf{W}_0\mathbf{x} + \frac{\alpha}{\sqrt{r}}\mathbf{B}\mathbf{A}\mathbf{x}$$

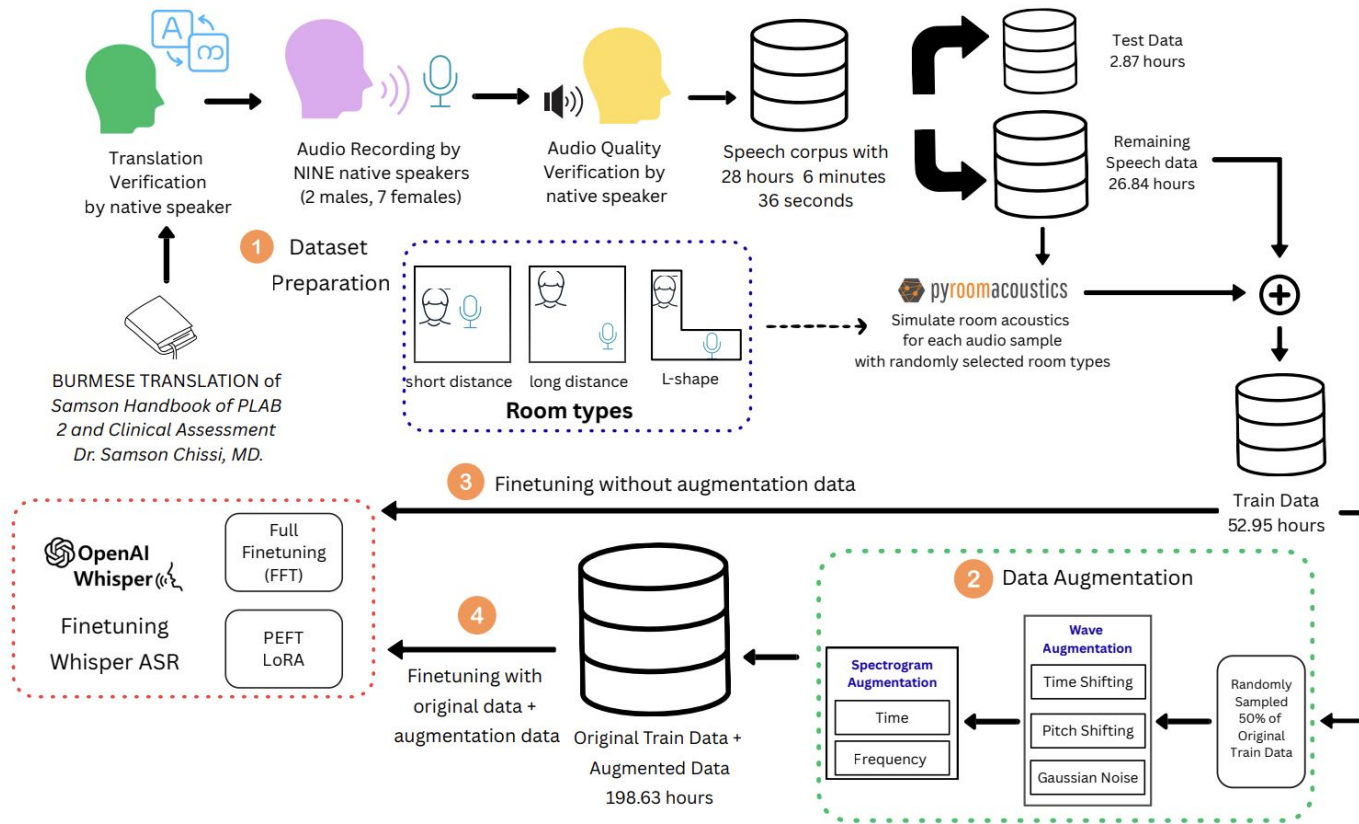
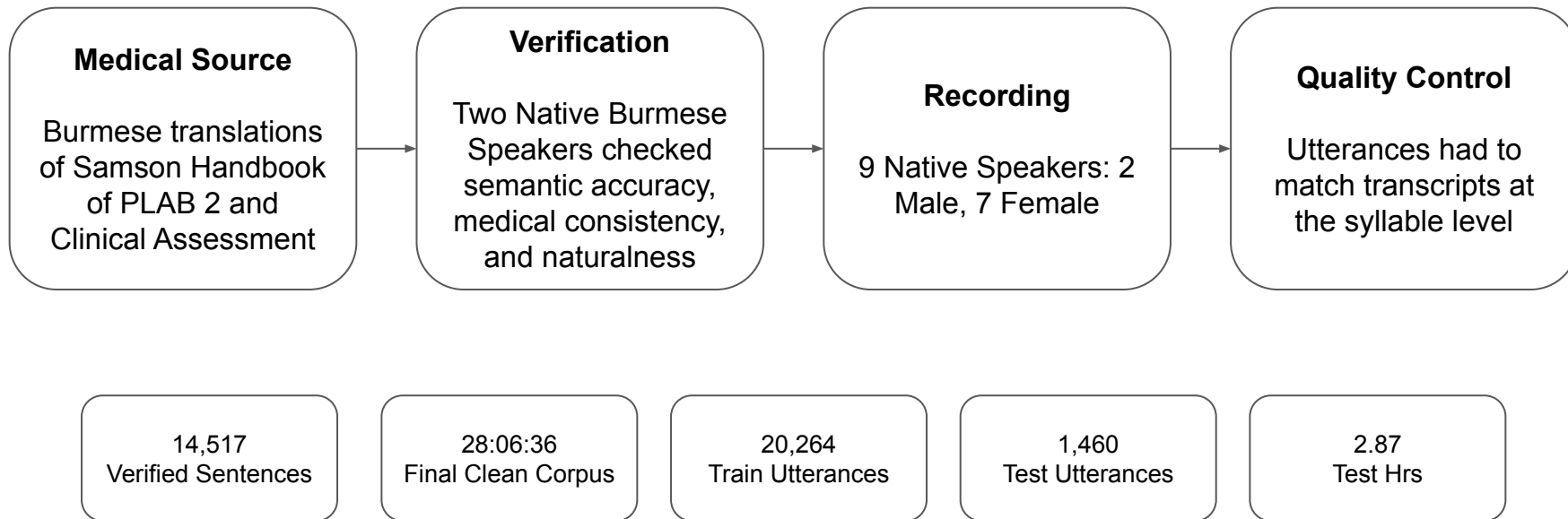


Figure 1: Pipeline for (1) dataset preparation, (2) augmentation, fine-tuning (3) without and (4) with augmentation

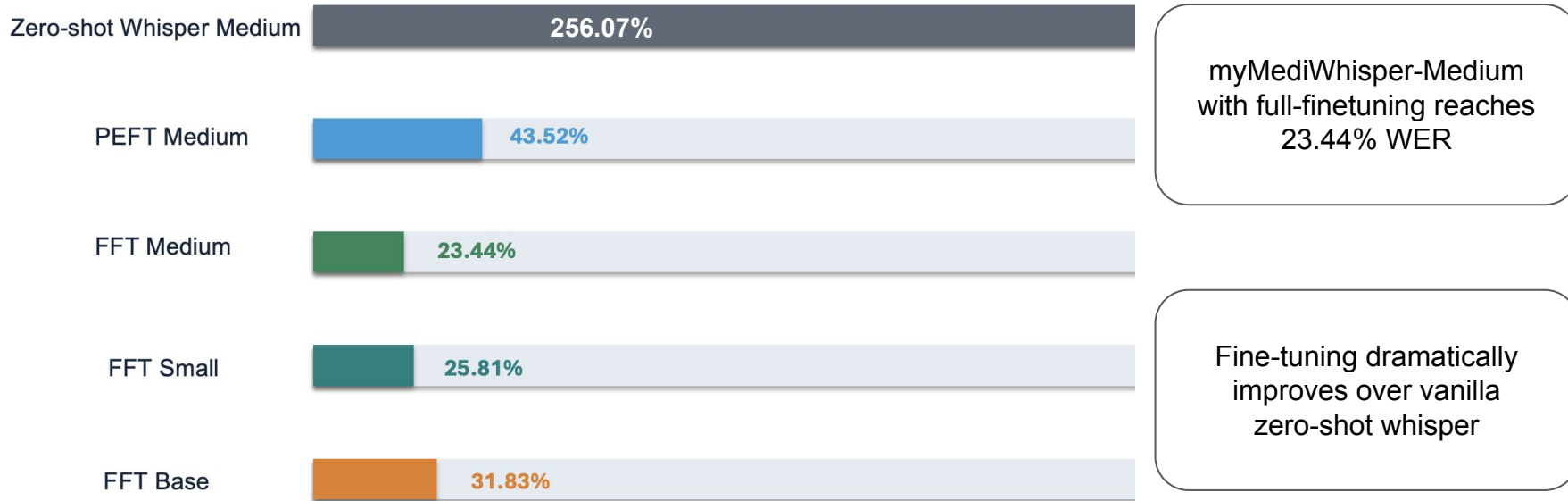
Dataset Construction



Research Questions

- How do full fine-tuning (FFT) and parameter-efficient fine-tuning (PEFT) compare for Burmese Medical ASR?
- How does Data Augmentation influence recognition accuracy and robustness?
- How well do models generalize to noise and room acoustic variations?

Results and Discussion



Evaluation

Substitution Errors

A word in the hypothesis is replaced with a different word from the reference

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ

ခေါင်းကိုက်

နေ

သည်

Deletion Errors

A word from the reference is missing in the hypothesis

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ



နေ

သည်

Insertion Errors

An extra word appears in the hypothesis that is not in the reference

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ

အတော်လေး

ဖျားနာ

နေ

သည်

Evaluation (Cont'd)

Word Errors

Overall measure of errors, which accounts for all SER, DER, and IER

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ

ဖျားနာ

နေ

သည်

$$WER = \frac{S + D + I}{N}$$

S: substitutions D: deletions I: insertions

N: total words in reference

Evaluation (Cont'd)



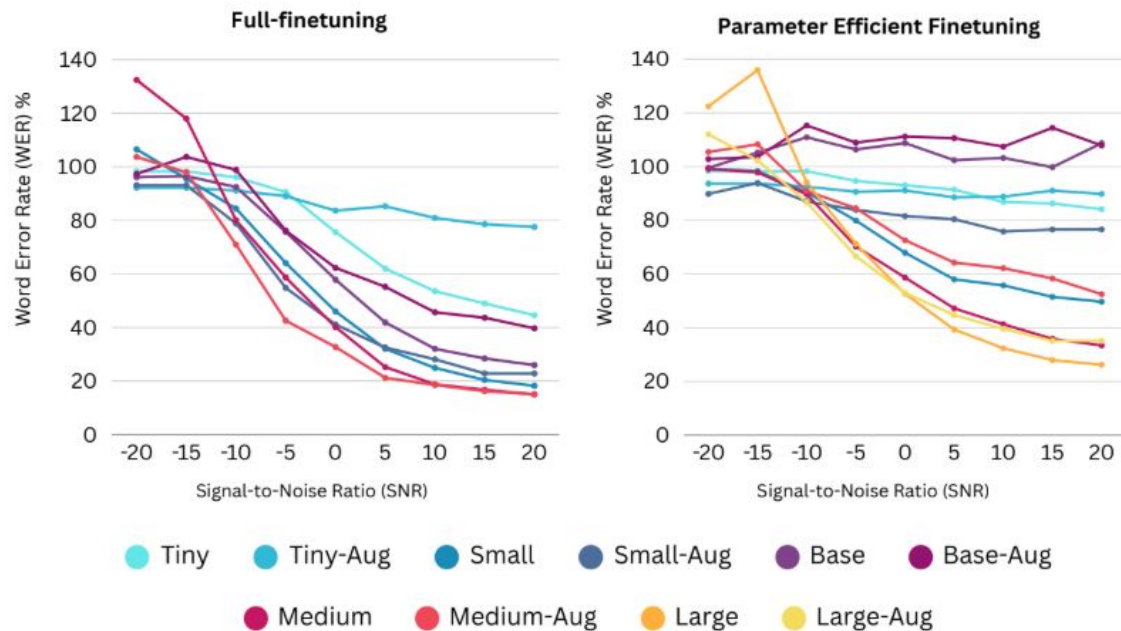


Figure 2: WER under varying Signal-to-Noise Ratio (SNR) conditions for myMediWhisper models trained with and without data augmentation



Figure 3: Effect of simulated room acoustics on WER for Whisper-Medium, comparing without and with augmentation across room types and speaker gender

Q & A